

## Rao-Cramér Lower Bound

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- For a sample of size one, say  $X_1$ , Fisher information is  $\text{Var}(\partial \log f(X_1; \theta) / \partial \theta)$ . What about a sample of size  $n$ ?
- Let  $X_1, \dots, X_n$  be a random sample from pdf  $f(x; \theta)$ . The information in the sample is

$$\text{Var} \left( \frac{\partial \log L(\theta)}{\partial \theta} \right) = nI(\theta),$$

where

$$\frac{\partial \log L(\theta)}{\partial \theta} = \sum_{i=1}^n \partial \log f(X_i; \theta) / \partial \theta.$$

Thus, the information in a sample of size  $n$  is  $n$  times the information in a sample of size one.

- Theorem 6.2.1 (Rao-Cramér Lower Bound): Let  $X_1, \dots, X_n$  be i.i.d. with common pdf  $f(x; \theta)$  for  $\theta \in \Omega$ . Assume that the regularity conditions R0-R4 hold. Let  $Y = u(X_1, \dots, X_n)$  be a statistics with mean  $\mathbb{E}(Y) = k(\theta)$ . Then,

$$\text{Var}(Y) \geq \frac{[k'(\theta)]^2}{nI(\theta)}.$$

- Corollary 6.2.1. Under the assumption of Theorem 6.2.1., if  $\mathbb{E}Y = \theta$ , then the Rao-Cramér Lower Bound becomes

$$\text{Var}(Y) \geq \frac{1}{nI(\theta)}.$$

- Consider the Bernoulli model with probability of success  $\theta$ . Then  $1/nI(\theta) = \theta(1 - \theta)/n$ . The mle of  $\theta$  is  $\bar{X}$  and  $Var(\bar{X}) = \theta(1 - \theta)/n$ . That is, in this case the variance of the mle has attained the Rao-Cramér Lower Bound.
- **Efficient Estimator**: Let  $Y$  be an unbiased estimator of  $\theta$ . The statistics  $Y$  is called an **Efficient Estimator** if and only if the variance of  $Y$  attains the Rao-Cramér Lower Bound.
- **Efficiency** of the estimator: the ratio of the Rao-Cramér Lower Bound to the actual variance of any unbiased estimation of a parameter.

- Poisson( $\theta$ ) distribution: Let  $X_1, \dots, X_n$  denote a random sample from Poisson distribution with mean  $\theta$ . It is known that  $\bar{X}$  is the mle of  $\theta$ . We also have that  $I(\theta) = 1/\theta$ . So the Rao-Cramér Lower Bound is  $\theta/n$ . But  $\theta/n$  is the variance of  $\bar{X}$ . Hence,  $\bar{X}$  is an efficient estimator of  $\theta$ .

- Beta( $\theta, 1$ ) distribution: Let  $X_1, \dots, X_n$  denote a random sample of size  $n > 2$  from distribution with pdf  $f(x; \theta) = \theta x^{\theta-1}$  for  $0 < x < 1$ , where  $\theta \in \Omega = (0, \infty)$ . The derivative of  $\log f$  is

$$\frac{\partial \log f}{\partial \theta} = \log x + \frac{1}{\theta}.$$

From this we have  $\partial^2 \log f / \partial \theta^2 = -\theta^{-2}$ . Hence  $I(\theta) = \theta^{-2}$ .

Next we find the mle. The log likelihood function

$$l(\theta) = \theta \sum_{i=1}^n \log x_i - \sum_{i=1}^n \log x_i + n \log \theta$$

The mle is  $\hat{\theta} = -n / \sum_{i=1}^n \log X_i$ . We can obtain

$$\text{Var}(\hat{\theta}) = \theta^2 \frac{n^2}{(n-1)^2(n-2)}.$$

It follows that this variance is larger than the Rao-Cramér Lower Bound, and hence,  $\hat{\theta}$  is not efficient. But notice that its efficiency converges to 1 as  $n \rightarrow \infty$ . Later, we will say that  $\hat{\theta}$  is asymptotically efficiently.